

Scaffold-redundant function

One Functional Specification S · many distinct full molecules · shared surrogate behavior

THE CLAIM

Behavior is scaffold-redundant under S: chemically different molecules realize the same Spec.

THIS NEIGHBORHOOD

59
unique Murcko in
S k=64 neighbors

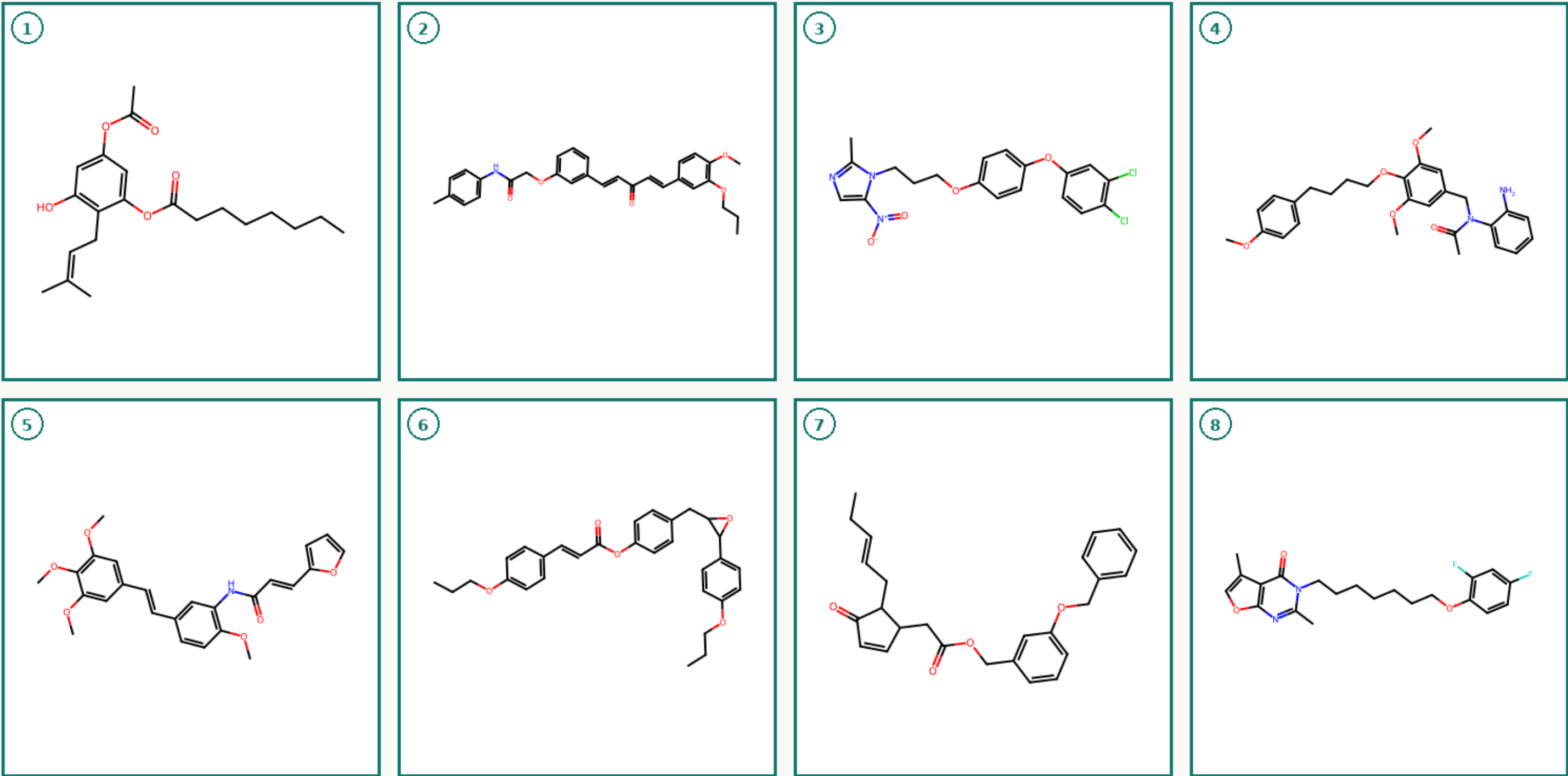
GENERATION (G5)

114
scaffolds from one Spec
(batch LogP std= 0.12)

ONE SPEC S

shared behavior

S neighbors — full molecules (unique Murcko), all map to the same Spec



Contrast: ECFP neighbors for the same probe (full molecules)

k=64: 46 unique scaffolds · higher v_beh (0.73 vs S=0.26) · structure search ≠ scaffold-redundant function

